

Derivation of Buoyancy Formula

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Assume that we have a rigid body Σ immersed into the liquid with density ρ . Then we partition the boundary of the body $\partial\Sigma$ into infinite surface elements dS , and also partition the whole body into volume elements dV .

For each surface element, the pressure can be expressed as:

$$p = p_0 + \rho gh$$

by the principle of Passca. Accordingly we can express the force:

$$dF = p dS$$

As the direction of the outer normal vector of surface element and the direction of the force is opposite, we have the vector form:

$$d\mathbf{F} = -p \cdot d\mathbf{S}$$

Then the total force can be calculated by integration:

$$F = \oint_{\partial\Sigma} dF = \oint_{\partial\Sigma} -p \cdot d\mathbf{S}$$

To make the integral match the vector form, we multiply a constant unit vector $\mathbf{c}$ in the integrand:

$$F = - \oint_{\partial\Sigma} p \cdot \mathbf{c} d\mathbf{S}$$

Then we consider the vector field $\mathbf{f} = p \cdot \mathbf{c}$, then use Gauss theorem:

$$F = -\Phi(\mathbf{f}, \partial\Sigma) = - \iiint_{\Sigma} \nabla \cdot \mathbf{f} dV$$

As the divergence:

$$\nabla \cdot \mathbf{f} = \nabla \cdot (p \cdot \mathbf{c}) = \nabla p \cdot \mathbf{c} + p(\nabla \cdot \mathbf{c}) = \nabla p \cdot \mathbf{c}$$

As the gradient of p can be computed:

$$\nabla p = (0, 0, \frac{\partial p}{\partial z}) = -\rho \mathbf{g}$$

We have:

$$F = - \iiint_{\Sigma} -\rho \mathbf{g} \cdot \mathbf{c} dV = \rho g \iiint_{\Sigma} dV = \rho g V$$